Mustafa Eren KOÇ

Software Engineer · Mobile, Backend & Data Products

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Summary

Computer Engineering graduate (ITU, 2026) who ships complete products: native iOS and Android apps, Python/FastAPI backends and the data pipelines behind them. Built **Klaris** (700,000+ products), **Hocam Kim?** (184,668 academics, about 4.9 million publications) and **TRBio** (two first-author manuscripts in preparation). Also brings two and a half years of part-time bank IT audit experience and a second B.S. in Business Administration.

Projects

Klaris — Founder & Software Engineer · klaris.com.tr — Mar 2026 – Present

- Designed, built and launched a cosmetic-ingredient app single-handedly (App Store and Google Play, September 2026): a barcode scan returns plain-Turkish ingredient explanations and a product score with open reasoning.
- Native iOS (Swift, SwiftUI, StoreKit) and Android (Kotlin, Jetpack Compose, ML Kit) apps on a FastAPI, PostgreSQL and Redis backend and a Python pipeline behind a 700,000+ product catalog; self-hosted with Docker, Nginx and CI/CD.
- Scoring engine with a product score and a personal match across four skin and four hair dimensions; 16,000+ automated tests across six codebases.

Hocam Kim? — Founder & Software Engineer · Independent Public Project · hocamkim.com — Oct 2026 – Present

- Built a public directory of 184,668 academics across 209 institutions with about 4.9 million deduplicated publications (Meilisearch), linking 111,000+ of them to OpenAlex via ORCID and shared DOIs.
- Designed an open, versioned scoring method with eight components, including SCImago Journal Rank (SJR) quartiles, citation percentiles, projects, theses, collaboration and patents, recomputed daily within each field; FastAPI, MongoDB, 3,800+ tests.

TRBio — Bioinformatics Portal of Türkiye — Co-Founder & Software Engineer · trbio.info — Sep 2025 – Present

- Built the identity-resolution, master-data, scoring and portal layers (Python, FastAPI, MongoDB, D3.js) linking national academic records to OpenAlex via ORCID.
- First author of two manuscripts in preparation (target venues: Briefings in Bioinformatics, Database).

Experience

IT Auditor (Part-Time) — Kuveyt Türk Participation Bank, Kocaeli — Jan 2024 – Jun 2026

- Audited IT general controls (access, change management, backup and business continuity) against BDDK (banking regulator) information-systems rules and KVKK (data-protection law).
- Authored a 26-page benchmark audit of the mobile banking apps of 10 leading Turkish banks across five areas (store ratings and releases, onboarding, login security, customer acquisition, money transfers).

AI Trainer (Coding) · Freelance — Outlier, Remote — Dec 2024 – Jun 2026

- Wrote coding prompts and reference solutions (Python, C/C++, JavaScript/TypeScript) and rated model responses (RLHF) to train LLMs.

Software Intern — Arçelik, Bolu — Aug – Sep 2023

- Built a full-stack web app (React, Node.js/Express, PostgreSQL) to replace Excel-based tracking of factory test data.

Network & Hardware Intern — Ministry of Youth and Sports of Türkiye, Ankara — Jul 2023

- Installed and configured Cisco routers and switches (IP addressing, subnetting, security settings); tested and troubleshot the network with Wireshark.

Education

B.S. in Computer Engineering (100% English) — Istanbul Technical University, Istanbul — Jul 2026
Kale Foundation (KSV) scholarship holder throughout the degree

B.S. in Business Administration — Anadolu University, Faculty of Business Administration, Eskişehir — May 2025
Graduated with Honors

Summer English Program — George Mason University, Virginia, USA — Summer 2019

Course Projects

LLM code/test generation on HumanEval-X (llama.cpp, JUnit, JaCoCo, PITest) · fully local RAG assistant (Ollama, ChromaDB, Streamlit) · concurrent crawler with live search (Python stdlib, SQLite) · trade-data ETL and Flask/MySQL analytics

Skills

Programming: Python, Swift, Kotlin, TypeScript/JavaScript, SQL · also Java, C/C++
Backend & Data: FastAPI, Flask, Node.js/Express, PostgreSQL, MongoDB, MySQL, Redis, Meilisearch, ETL pipelines, pytest
Mobile & Web: SwiftUI, UIKit, StoreKit, Jetpack Compose, Google Play Billing, ML Kit, Next.js, React, D3.js, Playwright
Infrastructure: Linux, Docker, Nginx, CI/CD, Cloudflare, systemd
AI & LLMs: LLM integration and evaluation, prompt engineering, RAG (ChromaDB), local models (Ollama, llama.cpp), RLHF
Languages: Turkish (native) · English (full professional proficiency; B.S. taught 100% in English)

Activities

ITU ACM Student Chapter, Events Team Member (2023–2026) · ITU Genç TEMA, Member (2021–2026)
Events: Audit & Advisory Fest (2025) · Start Camp, Özyeğin (2020) · ITU summits (2018–2020) · Leadership Summit, Boğaziçi (2017)
Work & Travel USA: retail in New Jersey (2021) and delivery in New York City (2022)